

Rail Fence:

- ① Encrypt the message "Enjoy your journey" using Rail Fence with depth 2 and depth 3:

Depth 2: Plain text: Enjoy your journey
Ciphertext?

E	J	Y	O	R	O	R	E
N		U		J		N	Y

Ciphertext: EJYORORENOYUJUNY

Depth 3:

E		Y		R		R
N	O	U	J	U	N	Y
	J		O		O	E

Ciphertext: EYRRNOYUJUNYJOOE

② Encrypt the message as per your choice using Rail Fence with depth 2 & 3.

Plaintext : Aansh Hegde
Aanshhegde

Depth 2:

A	N	H	E	D	
	A	S	H	G	E

Ciphertext: ANHEDASHGE

Depth 3:

A			H		D	
	A	S		H	G	E
		N		E		

Ciphertext: AHDASHGENE

- ③ Decrypt the Cipher "EOCSNYUWLEJTYREASON" encrypted using Rail Fence with depth 4:

E			O			C			S
	N		Y		U		W		L
		J		Y		R		E	
			O			N			S

Plain Text: "ENJOY YOUR NEW CLASSES"
 "Enjoy your new classes"

- ④ Decrypt the Cipher "BOOIGUYUBOSNUUTYRKAS" encrypted using Rail Fence with depth 3.

B		O		O		I		G	
	U		Y		U		B		O
		Y			R			K	
						S		N	
							U		U
								A	
									S

Plain Text: ~~BUY~~ BUY YOUR BOOKS IN AUGUST
 "Buy your books in August"

- ⑤ Decrypt the Cipher "SEMSERTISINCMOO" encrypted using Rail Fence with depth 2.

S		E		M		S		E		R		T		I
	S		I		N		E		M		O		O	

Plain Text: SSEIMNSCEMROTOI

Permutation

- ① Encrypt the following message "We are discovered" using the key ~~531~~ "53124".

Message: "We are discovered"
We are discovered

Key = 53124

①	②	③	④	⑤
5	3	1	2	4
w	e	a	r	e
d	i	s	c	o
v	e	r	e	d

Order: ③, ④, ②, ⑤, ①

a s r r c e e i e e o d w d v - CipherText.

DES

Initial Permutation:

58	50	42	34	26	18	10	02
60	52	44	36	28	20	12	04
62	54	46	38	30	22	14	06
64	56	48	40	32	24	16	08
57	49	41	33	25	17	09	01
59	51	43	35	27	19	11	03
61	53	45	37	29	21	13	05
63	55	47	39	31	23	15	07

- ① Using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as:

0X0002 0000 0000 0001

Converting them to the binary

2 - 0000

0000 0000 0000 0001 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0000 0001

1 - 0000

4 - 0100

Only two bits are '1': input bits 15 and 164
Every other are 0.

64 appears in Column 1, Row 4 → that is 25th entry in the table → Output bits 25 input bit 64 = 1

15 appears in Row 8, Column 7 → that is 63rd entry in the table → Output bits 63 = input bit 15 = 1

So other output bits = 0

0000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0000 0000 0000 0010 ,

Output:

0X0000 0010 0000 0002

- ② Given the 4 bit binary plaintext (M) 1101 and the key (K) 1011. Calculate the following resulting ciphertext (C) using a bitwise exclusive - OR (XOR) operation. Show your bit-by-bit work. Plaintext (M) = 1101 (binary), Key = 1011 (binary)

XOR of M = 1101 and K = 1011

$$\begin{array}{r} M = 1 \ 1 \ 0 \ 1 \\ K = 1 \ 0 \ 1 \ 1 \\ \hline 0 \ 1 \ 1 \ 0 \end{array}$$

C = 0110

- ③ The input to S-box 1 is 000000. What is the output?

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

input = 000000

Check the 0th position - 14

14 - 01110